

TRANSMITTAL	TRANSMITTAL NO. KC-RES-05-T01	DATE October 2026	SENT VIA Link · PDF attachment	PACKAGE 9 forms + appendix
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TO Hiring team · MEP / BIM group	FROM Kenneth Chow · Burnaby, BC · +1 778-791-3768 · wmchow5@gmail.com
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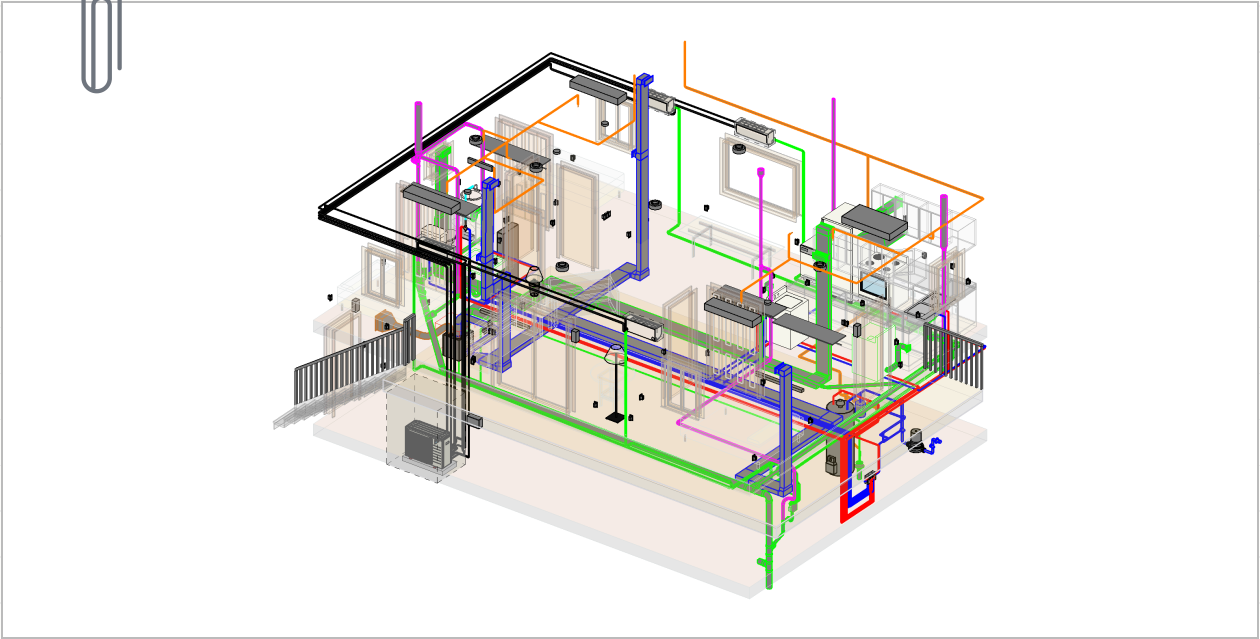
KENNETH CHOW

Re: MEP technician / MEP modeler roles, and coordinator roles where they fit. Enclosed is my BCIT residential MEP model, the QA review I ran on it against BC code, and the drawings.

ITEMS ENCLOSED	
ENCL.	DESCRIPTION
01	3D MEP coordination model, KC-RES-05 (this page)
02	Applicant qualifications and model register
03	Design review summary: 51 issues across seven disciplines
04	Five case studies with code basis, decision and before / after
05	Eight drawings exported from the live model
App. A	Full design review log, 51 items (separate PDF)

ENCL. 01 · 3D MEP coordination · view C-3D-OVERVIEW

Exported from Revit 2027, 2 Oct 2026



REMARKS

One-storey house on a basement, Burnaby BC. HVAC, plumbing, fire protection and electrical in one Revit model: 391 pipes (about 451 m), 26 circuits, 28 view templates, 8 Revit drawing sheets.

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APPLICANT QUALIFICATIONS		ENCL. 02	APPLICANT Kenneth Chow	BASED IN Burnaby, BC	
TRAINING Building Information Modeling (BIM) studies, BCIT, 2026. Bachelor of Engineering (Honours), Electronic and Communications, Hong Kong, 2015.	MODEL REGISTER · KC-RES-05				
	PIPES 391 · about 451 m				
	ELECTRICAL CIRCUITS 26 on panel P1 (200 A)				
EXPERIENCE Project Coordinator & Electronic Engineer (ELV systems), Shun Hing Engineering Contracting, Hong Kong, Nov 2020 - Sep 2023. Layout drawings, riser diagrams and shop drawings for access control, CCTV and intercom; RFIs, submittals and design markups; wall-cavity checks and power load calculations.	VIEW TEMPLATES / FILTERS 28 / 13				
	REVIT DRAWING SHEETS 8				
	ISSUES RAISED 51				
	CLOSED 40				
WHAT THAT MEANS HERE I have run the paperwork this page is built from. On KC-RES-05 I applied the same habit to my own model: raise the issue, cite the code, propose options, record the decision.	CLOSED, PLACEHOLDER FAMILY 6				
	KEPT, REASON RECORDED 3				
	DOCUMENTED 2				
CODES WORKED TO ON THIS MODEL BC Building Code 2024 Part 9 · BC Plumbing Code · Canadian Electrical Code (as adopted in BC) · NFPA 13D · CSA F280, F326, B52 · CAN/ULC-S553 · Burnaby waterworks bylaw.	OPEN 0				
	BACKUPS BEFORE EACH CHANGE one per item				
SOFTWARE Autodesk Revit (MEP: piping, ducts, circuits, panel schedules, view templates, families) · AutoCAD · Revit API checks through pyRevit.	DESIGN BASIS Equipment selections, fixture locations and sprinkler design flows follow the BCIT project brief. Where the code review called for a change, such as heat-pump indoor units to meet the BCBC heating requirement (M-09), the change and its reasoning are recorded in the design review log.				
METHOD I ran the geometry and code checks through the Revit API (pyRevit), with Claude as a scripting assistant. I set the rules, made each decision and verified every change in the model.					

DESIGN REVIEW SUMMARY			ENCL. 03	PROJECT KC-RES-05	REVIEWED BY K. Chow	PERIOD 28 Sep - 6 Oct 2026
SUMMARY I reviewed my BCIT residential MEP model against BC code before issue. 51 issues across seven disciplines: 40 closed, 6 closed with placeholder families pending manufacturer content, 3 kept by design with reasons, 2 documented, none open. Every change was backed up, clash-checked and logged.						
ISSUES BY DISCIPLINE AND STATUS				QA PROCESS, PER ISSUE		
DISCIPLINE	CLOSED	PLACEHOLDER FAMILY	KEPT	DOCUMENTED	OPEN	TOTAL
HVAC	12	3	-	-	-	15
Plumbing	7	2	1	-	-	10
Electrical	7	-	-	-	-	7
Fire protection	3	1	1	1	-	6
Architectural	4	-	-	1	-	5
Fire alarm	4	-	-	-	-	4
BIM standards	3	-	1	-	-	4
Total	40	6	3	2	0	51
				MINOR FIXES		
				System classifications, level names, imperial family names and window type names corrected in one pass (P-06, P-23, P-24, P-25).		
				FULL LOG		
				All 51 items, with code basis and status, are in <u>Appendix A</u> .		

DESIGN REVIEW RECORD

ITEM NO.
C-03

ENCL.
04-1

DISCIPLINE
Fire protection

STATUS
Closed

SUBJECT **Fire sprinkler main fed straight off domestic water**

ISSUE

The 1¼" sprinkler main teed off the domestic cold water through a transition fitting. No backflow preventer, control valve, gauge or drain & test.

OPTIONS CONSIDERED

1. Add the standard 13D riser components. 2. Treat it as a multipurpose system. Rejected: the sprinkler piping dead-ends, so water never circulates.

RESOLUTION

Five in-line placeholders on the fire main in the basement, with the drain & test discharging over the floor drain, in the order shown under After.

CODE BASIS

NFPA 13D riser components and waterflow alarm · BC Plumbing Code backflow, CSA B64.10 (type set by the City).

DECISION AND REASONING

I placed the flow switch upstream of the drain & test valve. A flow test has to move water past the switch, or it can't prove the alarm works. Tracing where the test water goes showed the first order was wrong, so I swapped the two.

REFERENCE

FP-CV1 1774562 · FP-BFP1 1774565 · FP-PG1 1774568 · FP-FS1 1774574 · FP-DT1 1795087

BEFORE

DCW tee → fire main → sprinklers

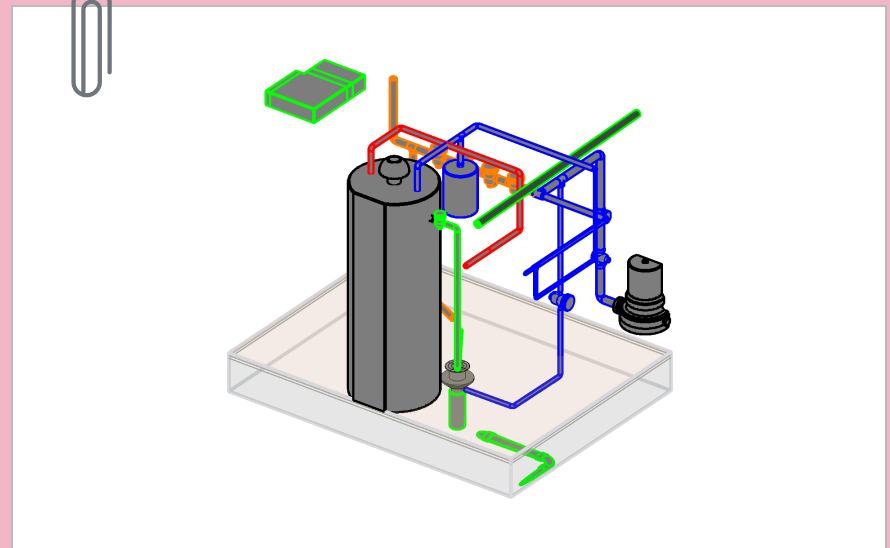
No way to isolate, test, or stop stagnant sprinkler water flowing back into drinking water.

First build order: CV → BFP → PG → D&T → FS. A flow test would never pass the switch.

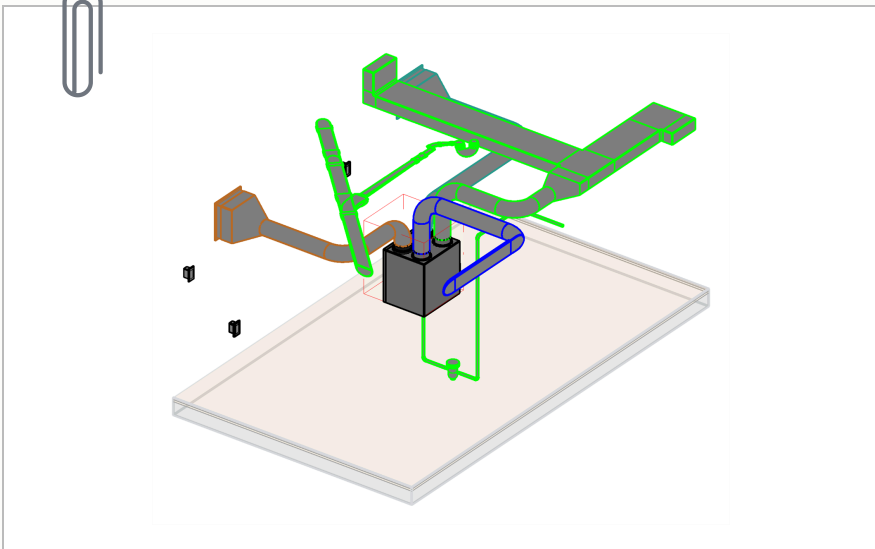
AFTER

DCW tee → CV → BFP → PG → FS → tee → up

The drain & test sits on a 1" branch off a tee just past the flow switch. Opening it sends water through the switch first, so the test proves the alarm.



PINK - FILE COPY

DESIGN REVIEW RECORD		ITEM NO. P-09	ENCL. 04-2	DISCIPLINE HVAC	STATUS Closed
SUBJECT HRV exhaust discharging under the deck					
ISSUE The HRV exhaust grille sat under the deck, 0.53 m from the outdoor heat pump's clearance zone.			BEFORE Exhaust under the deck, 0.53 m from the heat-pump clearance zone. Moist air on deck joists, frost on the coil. Outdoor ducting about 4.5 m. No receptacle and no condensate drain at the HRV. AFTER Grilles on the west wall, 2.47 m apart; discharge about 3.5 m from the heat pump. Outdoor ducting about 3.0 m straight. Dedicated receptacle REC-HRV; condensate pump CP-01 to the receptor with a 26 mm air gap. 		
OPTIONS CONSIDERED Keep it · slide it east on the south wall · run a long duct to the west wall · rotate the HRV · rotate and move the HRV. Compared in one table.					
RESOLUTION Rotated HRV-01 90° and moved it beside the west wall: both grilles on the west wall, 2.47 m apart, out from under the deck. Then added the receptacle and condensate pump it was missing.					
CODE BASIS HRV manufacturer separation (≥ 1.8 m) and height above grade (≥ 450 mm); keep moist exhaust off deck framing and the heat-pump coil.					
DECISION AND REASONING I rotated the HRV 90° so both outdoor ports face the west wall. That took the exhaust out from under the deck, set the grilles 2.47 m apart and cut outdoor duct from about 4.5 m to 3.0 m. I compared five layouts first.					
REFERENCE HRV-01 1547124 · AT-10 1567942 · AT-17 1785468 · REC-HRV 1785491 · CP-01 1788255					

DESIGN REVIEW RECORD

ITEM NO.
M-09 · M-10

ENCL.
04-3

DISCIPLINE
HVAC

STATUS
Closed

SUBJECT **Heat-pump sizing and the R32 charge limit**

ISSUE

No CSA F280 calculation. The living room's 2.0 kW head served a 35 m² room with a 5.9 m vault. The heads were cooling-only, so the house had no real heat.

OPTIONS CONSIDERED

A. Heat-pump FTX heads on a heat-pump outdoor unit. B. Keep the coursework cooling-only heads and heat with about 2.3 kW of extra baseboard, which pushes the panel past 100 A.

RESOLUTION

Option A, recorded as a deviation from the BCIT equipment spec (FTC15NV14 × 3 + FTC20NV14). Then checked the R32 charge against the smallest room.

CODE BASIS

BCBC 9.33.3.1 heating (22 °C) sized to CSA F280 · CSA B52 / IEC 60335-2-40 charge limit for A2L refrigerant.

DECISION AND REASONING

I chose heat-pump heads over adding 2.3 kW of baseboard, knowing it departs from the coursework equipment. One system heats and cools, the panel stays within its rating, and heating uses about a third of the electricity. I then checked the R32 charge against the smallest room, because a multi-split can leak its whole charge into one bedroom.

REFERENCE

AC-01 1524859 · AC-02 1525077 · AC-03 1525454 · AC-04 1526631 · CU-01 1527078

BEFORE

AC-04 FTC20: **2.0 kW** against an open-zone cooling load of **3.3 kW**.

All four heads FTC, cooling only. Heat came from the backup baseboards added in C-01.

AFTER

F280-STYLE ESTIMATE house 3.6 kW heat / 4.6 kW cool

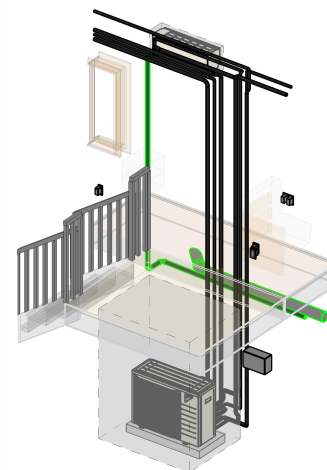
HEADS FTX15 × 3 + FTX35

LIQUID LINES 18.9 + 5.8 + 8.9 + 22.3 = 55.9 m

CHARGE 2.0 + (55.9 - 30) × 0.02 = **2.5 kg**

m_{max} (Bedroom 1) 2.5 × 0.307^{1.25} × 1.8 × √7.44 = **2.80 kg**

RESULT 2.5 ≤ 2.80, passes with no mitigation



DESIGN REVIEW RECORD

ITEM NO.
M-12

ENCL.
04-5

DISCIPLINE
Plumbing

STATUS
Closed

SUBJECT **A pipe through the water heater, and no floor drain**

ISSUE

A 1" PVC pipe hung off the water heater's drain-valve connector with its centreline inside the tank wall. Its comment said it was the T&P discharge. The basement had no floor drain for it.

OPTIONS CONSIDERED

(a) Extend the building drain stack down and drain a floor drain by gravity. (b) A sump and pump.

RESOLUTION

Deleted the pipe; added a real T&P valve with a $\frac{3}{4}$ " copper discharge; floor drain FD-01 with a P-trap, 3" drain at 2 % to a new sanitary tee; stack extended from -2.19 to -2.95 m.

CODE BASIS

BC Plumbing Code: T&P discharge full size and temperature-rated, no valves; every fixture trapped, seal ≥ 38 mm; floor drains need trap seal protection.

DECISION AND REASONING

I removed a PVC relief pipe that ran through the tank wall and gave the T&P valve a full-size copper discharge to a new trapped floor drain. I extended the building drain stack so the drain works by gravity, without a sump pump.

REFERENCE

FD-01 1782449 · trap 1783819 · tee 1782637 · TPV-01 1782320 · WH-01 1602886

BEFORE

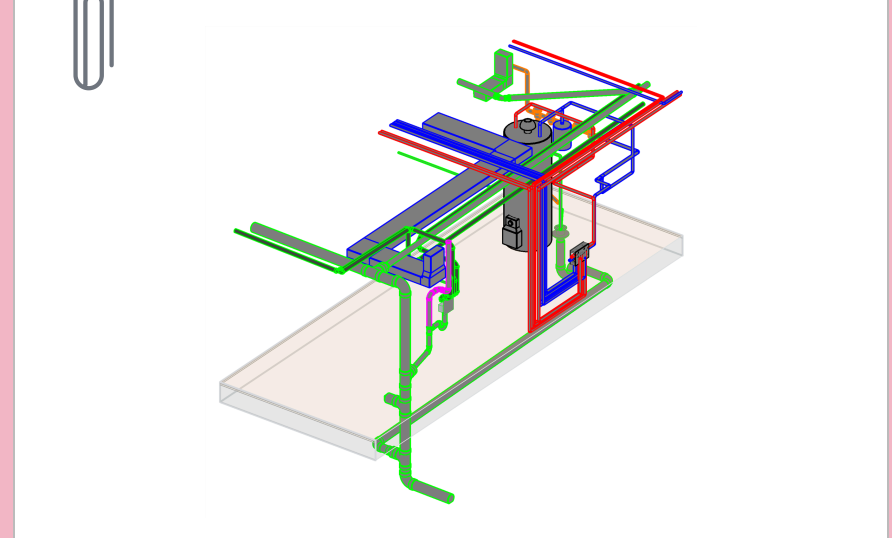
1" PVC on the tank's **drain valve**, clipping the tank. PVC is not rated for 99 °C relief water.

No floor drain; building drain exit only 0.14 m below the floor.

AFTER

WH-01 → TPV-01 → $\frac{3}{4}$ " Cu → air gap → FD-01 → P-trap → stack

Trap seal about 95 mm, kept primed by trap primer TP-01 on the cold-water main.



DRAWING ENCLOSURES

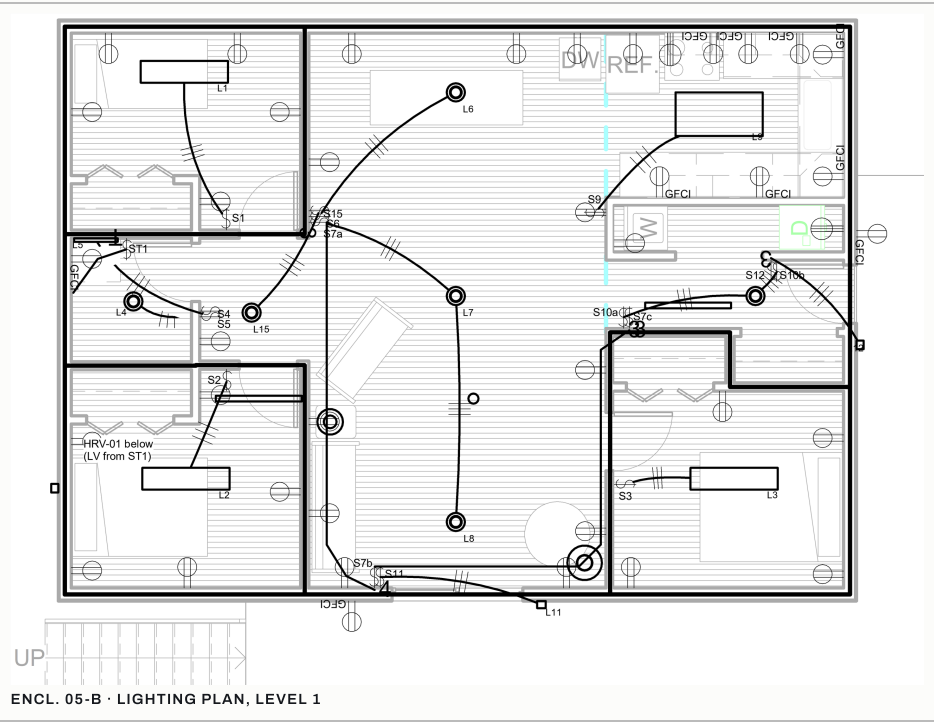
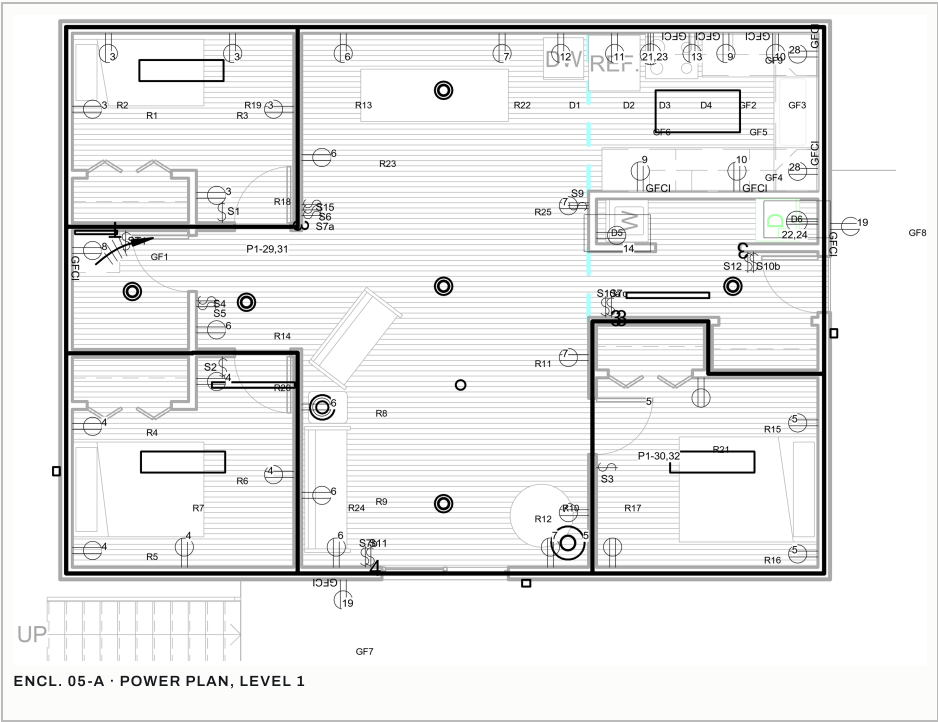
ENCL.
05 A-H

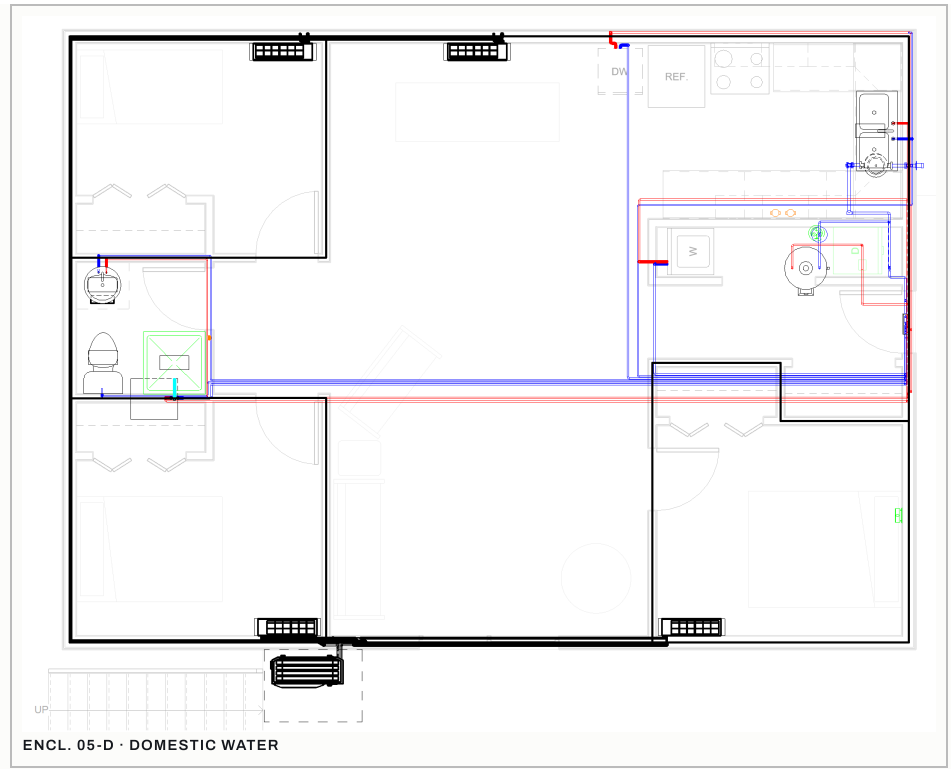
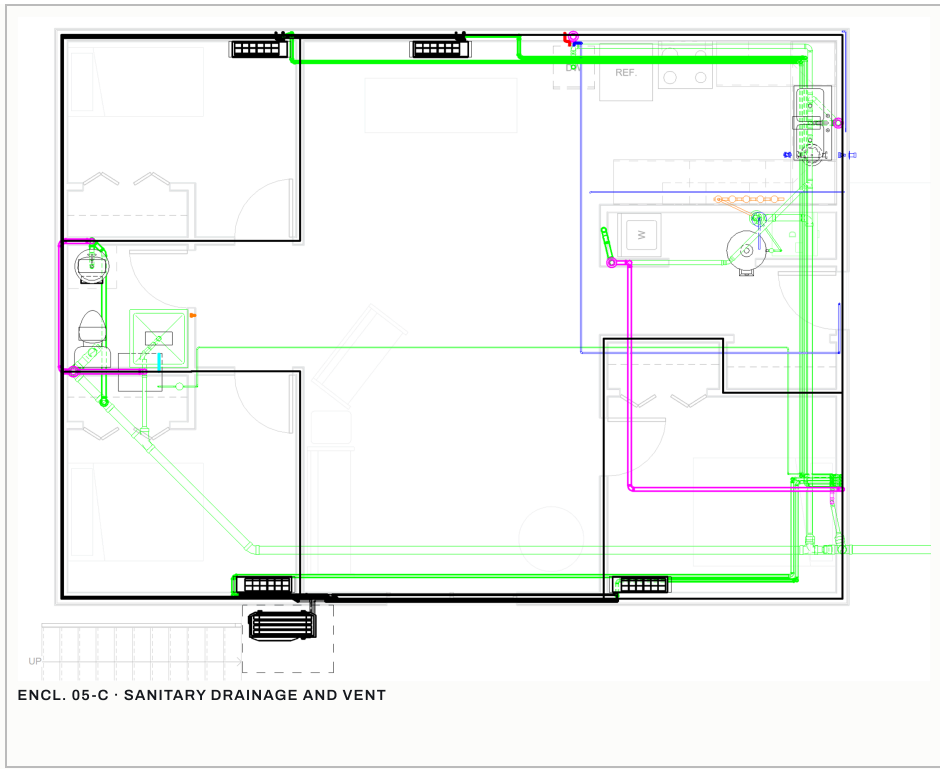
SOURCE
Revit views, exported 2
Oct 2026

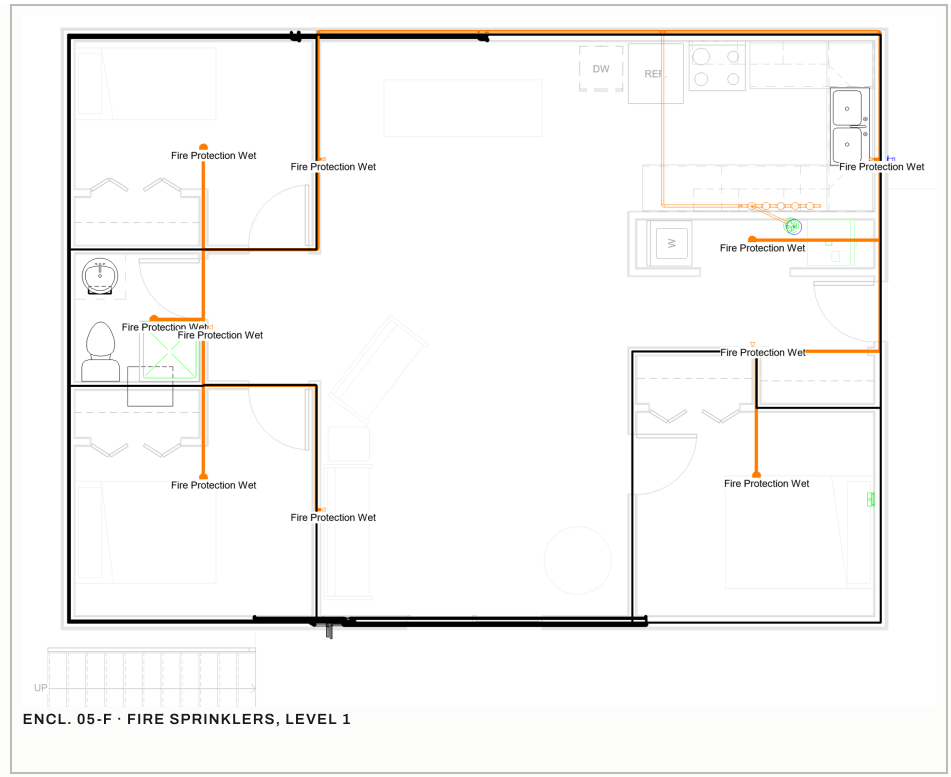
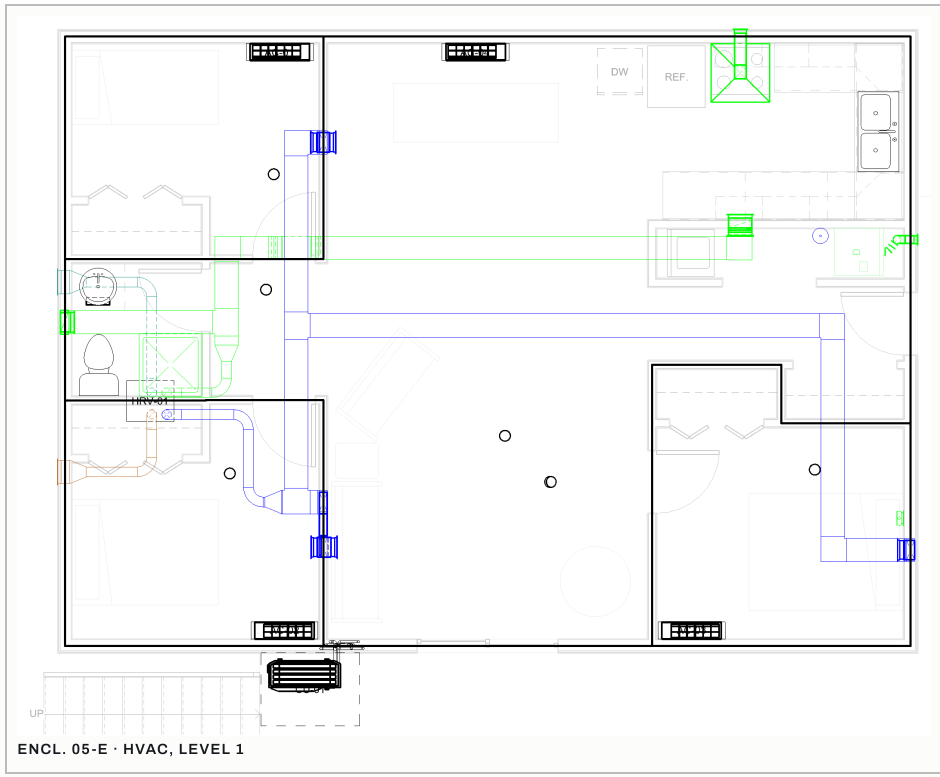
NOTE
Title blocks cropped

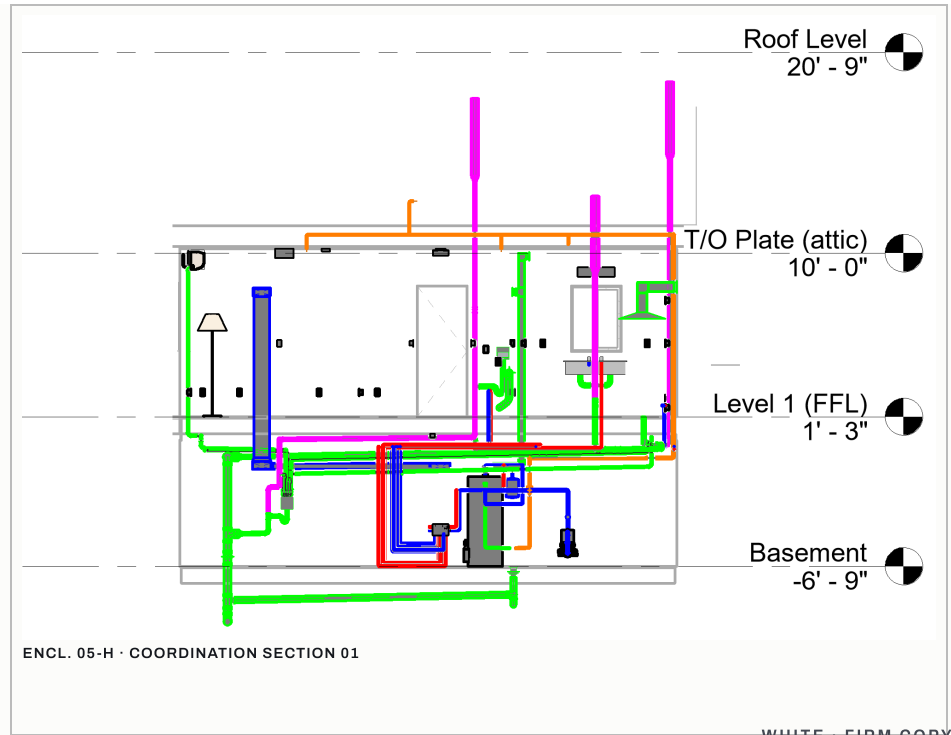
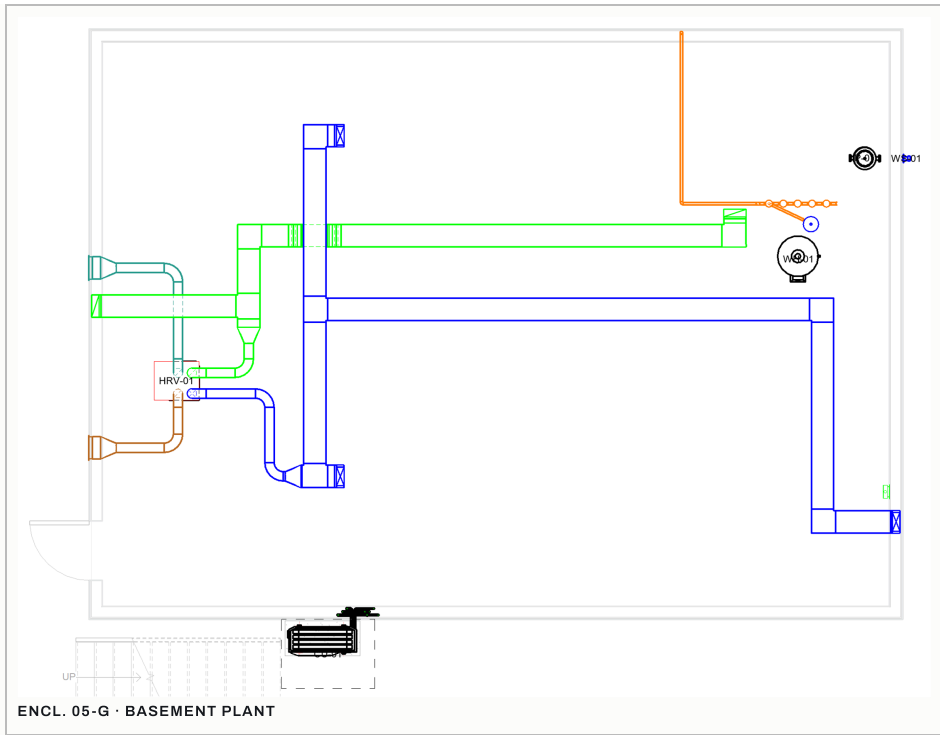
ENCL. 05-A · Power plan, Level 1

[Click the drawing to enlarge](#)









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REVIEWER ACTION

TRANSMITTAL NO.
KC-RES-05-T01

RETURN TO
Kenneth Chow

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BASED IN **Burnaby, BC**

OPEN ITEMS

None. R-01, raised by my own second review (smoke alarm SD1 0.64 m from supply diffuser AT-06), is closed: SD1 now sits at least 1.8 m from all supply air. The first spot I proposed failed against the AC-01 head, so I searched the room for a better one.

SECOND-PASS MODEL RE-CHECK

Nine further items checked in the model and closed: AC heads against partitions, the drain & test branch, pendant stems, 2" drain slopes, unlevelled parts, basement heat, the dryer vent (moved to the roof), outdoor-unit clearance under the deck and a trap primer for FD-01.

Placeholder families (heat pump, valves, shower valve, range hood) stand in for manufacturer families until product selection.

PREPARED BY

Kenneth Chow · BIM & design coordination · BCIT BIM studies 2026.

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